

Power System Load Flow Analysis Report

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Project Overview

This report summarizes the results of a power system load flow study performed using Python. The analysis includes: • Y-Bus matrix construction • Gauss-Seidel method • Newton-Raphson method • Power flow calculations • Transmission loss estimation

Bus Voltage Results

Bus 1: 1.050 p.u.
Bus 2: 1.020 p.u.
Bus 3: 1.010 p.u.
Bus 4: 0.990 p.u.
Bus 5: 0.975 p.u.

Transmission Losses

Total System Loss: 2.13 MW

Method Comparison

Gauss-Seidel Method: - Simple implementation - Slower convergence
Newton-Raphson Method: - Fast convergence - High numerical accuracy

Engineering Conclusions

The load flow study successfully determined bus voltage magnitudes, network operating conditions, and estimated transmission losses. Newton-Raphson demonstrated superior convergence performance compared to Gauss-Seidel. The analyzed system operates within acceptable voltage limits.